Abstract

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. Early detection and accurate diagnosis are essential for effective treatment and improving survival rates. Machine learning techniques can assist medical professionals in identifying cancer at an early stage by analyzing medical data. This project focuses on predicting breast cancer using an Artificial Neural Network (ANN) model.

The dataset used in this study is the Breast Cancer Wisconsin Dataset, which is available through the scikit-learn library. The dataset contains various features related to tumor characteristics such as radius, texture, perimeter, smoothness, and fractal dimension. These features are used as input to train the machine learning model.

In this project, the dataset is preprocessed and divided into training and testing sets. Feature scaling is applied to normalize the data, and an Artificial Neural Network model is implemented using the MLPClassifier algorithm. The trained model predicts whether a tumor is malignant (cancerous) or benign (non-cancerous). The performance of the model is evaluated using metrics such as accuracy, F1-score, and confusion matrix.

The results show that the ANN model achieves high accuracy in classifying breast cancer cases. This demonstrates that machine learning can be a useful tool in assisting medical diagnosis and supporting healthcare professionals in early detection and decision-making.

Aim / Objectives

Aim

The main aim of this project is to develop a machine learning model to predict whether a breast tumor is malignant or benign using medical data from the Breast Cancer Wisconsin Dataset with the help of an Artificial Neural Network implemented in scikit-learn.

Objectives

1. To analyze the breast cancer dataset and understand the important tumor features.
2. To preprocess the dataset by performing data cleaning and feature scaling.
3. To divide the dataset into training and testing sets for model development.
4. To implement an Artificial Neural Network (ANN) model using MLPClassifier.
5. To train the model to classify tumors as malignant or benign.
6. To evaluate the performance of the model using accuracy score, F1-score, and confusion matrix.
7. To demonstrate how machine learning can assist in early detection of breast cancer.

Introduction

Breast cancer is one of the most prevalent types of cancer affecting women globally and is a leading cause of cancer-related mortality. Early detection is critical because it significantly increases the chances of successful treatment and survival. Traditional diagnostic methods, such as mammography, ultrasound, biopsy, and histopathological analysis, while effective, have limitations. They are often time-consuming, costly, and subject to human interpretation errors.

To overcome these challenges, Machine Learning (ML) techniques have emerged as effective tools in medical diagnostics. ML models can analyze historical clinical data to detect complex patterns that may not be easily recognizable by humans. By leveraging patient and tumor features, ML algorithms can automatically classify tumors as malignant or benign, thereby assisting doctors in making faster and more accurate decisions

Dataset Overview

The dataset used in this project is the Breast Cancer Wisconsin (Diagnostic) Dataset provided by the scikit-learn library.

- Number of samples: 569
- Number of features: 30 numeric features representing tumor characteristics
- Target variable: 0 = Malignant, 1 = Benign
- Features include: radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, fractal dimension

This dataset contains quantitative measurements of cell nuclei from breast mass images. It is widely used in ML research for classification tasks due to its clean, well-structured, and labeled data.

Theory

The predictive model in this project is based on Artificial Neural Networks (ANNs), specifically the Multi-Layer Perceptron (MLP).

1. **Artificial Neural Network (ANN)**

ANNs are computational models inspired by the structure and function of the human brain. They consist of layers of interconnected nodes called neurons. Each neuron computes a weighted sum of its inputs, adds a bias term, and applies an activation function to introduce non-linearity. This allows ANNs to learn complex patterns in the data that linear models cannot capture.

2. **Multi-Layer Perceptron (MLP)**

MLPs are a type of feedforward neural network with one or more hidden layers. In this project, we use a single hidden layer with 100 neurons. The network is trained using backpropagation, which consists of:

- **Forward Pass:** Input data is passed through the network to generate predictions.
- **Error Calculation:** The difference between predicted and actual output is calculated.
- **Backward Pass:** Gradients of the error with respect to the network weights are computed, and weights are updated using gradient descent to minimize the error.

This iterative process continues until the network achieves optimal weights for accurate predictions.

3. **Activation Functions**

Activation functions introduce non-linear transformations, enabling the network to model complex relationships between input features and the output. Common activation functions include:

- **ReLU (Rectified Linear Unit):** Efficient for hidden layers, allows faster convergence.
- **Sigmoid:** Maps output to a range between 0 and 1, often used in binary classification.

4. **Feature Standardization**

ANNs are sensitive to feature scaling. All input features are standardized using StandardScaler, transforming them to have a mean of 0 and standard deviation of 1. This improves training stability and accelerates convergence.

Significance of the Project

- **Early Detection:** Enables timely intervention, potentially saving lives.
- **Enhanced Accuracy:** Reduces human error in medical diagnostics.
- **Automation:** Provides a framework for automated tumor classification using ML.
- **Research Foundation:** Serves as a basis for further exploration of deep learning models in healthcare.

Prerequisites / Requirements

Software Requirements

To implement this mini project, the following software tools are required:

- **Python 3.x:** A high-level programming language used for data analysis and machine learning. Python provides libraries that simplify ML implementation.
- **Jupyter Notebook / IDE:** An interactive environment such as **Jupyter Notebook**, **VS Code**, or **PyCharm** is recommended for writing and executing code. Jupyter allows step-by-step execution and visualization of outputs, which is helpful for project reports.

Python Libraries

The project relies on several Python libraries for data handling, visualization, machine learning, and evaluation:

1. **NumPy** – For numerical computations and handling multidimensional arrays.
2. **Pandas** – For data manipulation, creating dataframes, and easy exploration of datasets.
3. **Matplotlib** – For basic data visualization (line plots, scatter plots, etc.).
4. **Seaborn** – For advanced visualization, including heatmaps and correlation plots.
5. **Scikit-learn (sklearn)** – For implementing machine learning algorithms, including:
    - MLPClassifier for building the ANN model

- train_test_split for splitting dataset into training and testing sets
- StandardScaler for feature standardization
- accuracy_score, f1_score, and confusion_matrix for evaluating model performance

Hardware Requirements

- **Processor:** Minimum Intel i3 or equivalent
- **RAM:** 4 GB or more (8 GB recommended for faster computation)
- **Storage:** At least 2 GB free for Python and libraries

ANN training in this project is lightweight and can run efficiently on most modern laptops without requiring GPU acceleration

Dataset Requirements

- The project uses the Breast Cancer Wisconsin (Diagnostic) Dataset, which is included in scikit-learn:
  - 569 samples with 30 numeric features each
  - Target variable: 0 = Malignant, 1 = Benign
- The dataset should be loaded from sklearn or downloaded from the UCI Machine Learning Repository if needed.

Methodology / System Architecture

The methodology of the breast cancer prediction system describes the steps used to build the predictive model using machine learning techniques. The system analyzes medical data and classifies tumors as benign or malignant. The overall architecture consists of several stages, starting from data collection to model evaluation.

1. **Data Collection**

The dataset was obtained from the Breast Cancer Wisconsin Dataset available in the scikit-learn library. It contains medical measurements of breast tumors that help in identifying whether a tumor is malignant or benign.

2. **Data Preprocessing**

Data preprocessing is an important step to prepare the dataset for machine learning. In this step:

- The dataset is loaded into Python.
- Features and target variables are separated.
- The data is converted into a structured format using a DataFrame for easier analysis.

3. **Data Splitting**

The dataset is divided into two parts:

- Training Data (80%) – used to train the machine learning model.
- Testing Data (20%) – used to evaluate the performance of the model.

This helps in checking how well the model performs on unseen data.

4. **Feature Scaling**

Feature scaling is applied using the StandardScaler method. This process standardizes the data so that all features have a similar scale, which improves the performance and convergence of the neural network model.

5. **Model Development**

In this project, an Artificial Neural Network is implemented using the MLPClassifier algorithm from the scikit-learn library. The neural network consists of:

- An input layer
- One hidden layer with 100 neurons
- An output layer for classification

6. **Model Training**

The neural network model is trained using the training dataset. During training, the model learns the patterns and relationships between tumor features and their corresponding classes.

7. **Prediction**

After training, the model predicts whether a tumor is malignant or benign using the testing dataset.

8. **Model Evaluation**

The performance of the model is evaluated using different metrics such as:

- Accuracy Score – measures how many predictions are correct.
- F1 Score – evaluates the balance between precision and recall.
- Confusion Matrix – shows the number of correct and incorrect predictions.

These evaluation metrics help determine the effectiveness of the machine learning model in predicting breast cancer.

9. Overall Workflow

The overall workflow of the methodology can be summarized as follows:

1. Load the dataset
2. Explore and understand the data
3. Preprocess and scale the features
4. Split the dataset into training and testing sets
5. Build the Artificial Neural Network model
6. Train the model using training data
7. Predict tumor classes using test data
8. Evaluate the model performance using different metrics

Machine Learning Technique

In this project, the machine learning technique used for predicting breast cancer is an Artificial Neural Network (ANN). The model is implemented using the scikit-learn and trained using the Breast Cancer Wisconsin Dataset. Artificial Neural Networks are powerful machine learning models that are widely used for classification, prediction, and pattern recognition problems, especially in medical diagnosis and healthcare analytics.

Artificial Neural Network (ANN)

An Artificial Neural Network is a machine learning model inspired by the structure and functioning of the human brain. It consists of interconnected nodes called neurons that work together to process information and identify patterns in data. ANN models are widely used for classification, prediction, and pattern recognition tasks.

A neural network is typically composed of three main layers:

1. **Input Layer**

   The input layer receives the dataset features as input. In this project, the input layer contains 30 neurons corresponding to the 30 tumor features in the dataset.

2. **Hidden Layer**

   The hidden layer performs complex computations on the input data. In this project, one hidden layer with 100 neurons is used. Each neuron applies mathematical operations and activation functions to learn patterns in the data.

3. **Output Layer**

   The output layer produces the final prediction. Since this is a binary classification problem, the output layer predicts whether the tumor is malignant or benign.

MLPClassifier Algorithm

The ANN model in this project is implemented using the MLPClassifier (Multi-Layer Perceptron Classifier) available in scikit-learn. The MLPClassifier is a feedforward neural network that uses backpropagation to train the model.

Key parameters used in the model include:

- Hidden Layer Size: 100 neurons in one hidden layer
- Maximum Iterations: 500 training iterations
- Random State**:** 42 for reproducibility

During training, the neural network adjusts its weights and biases to minimize prediction error. The learning process continues until the model reaches optimal performance or the maximum number of iterations is completed.

Learning Process

The learning process of the neural network occurs in several stages:

1. The input data is passed to the input layer.
2. The input values are multiplied by weights and passed through the hidden layer.
3. Activation functions are applied to introduce non-linearity.
4. The output layer produces a prediction.
5. The predicted output is compared with the actual target value.

6. The error is calculated and propagated back through the network.
7. The weights and biases are updated to reduce the prediction error.

This process is repeated for multiple iterations until the model achieves optimal performance.

Advantages of Using ANN

- Capable of learning complex patterns in data
- Provides high accuracy in classification problems
- Suitable for medical data analysis and disease prediction
- Can handle large datasets with multiple features

Application in Breast Cancer Prediction

In this project, the ANN model learns the relationship between tumor features and their corresponding classifications. After training, the model can analyze new tumor data and predict whether it is malignant or benign. This helps demonstrate how machine learning techniques can assist healthcare professionals in early detection and diagnosis of breast cancer.

Implementation

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix,
classification_report

data = load_breast_cancer()
X = data.data
y = data.target
print("Shape of data:", X.shape)
print("Target names:", data.target_names)

df = pd.DataFrame(X, columns=data.feature_names)
df['target'] = y

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(100,),  # One hidden layer with 100 neurons
    max_iter=500,
    random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(len(X_test))

sample_index = 113
sample = X_test[sample_index].reshape(1, -1)

prediction = model.predict(sample)

if prediction[0] == 0:
    print(f"Prediction for sample {sample_index}: Malignant")
else:
    print(f"Prediction for sample {sample_index}: Benign")
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
f1 = f1_score(y_test, y_pred, average=None)
print("\nF1 Score for Malignant (0):", f1[0])
print("F1 Score for Benign (1):", f1[1])

print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))
```

Results

Single Sample Prediction

Test Sample Index: 113

```
Prediction for sample 113: Malignant
```

This demonstrates the model's ability to predict the tumor type for an unseen individual sample, simulating real-world clinical use.

Model Accuracy and F1-Score

```
Accuracy: 0.9736842105263158

F1 Score for Malignant (0): 0.9647058823529412
F1 Score for Benign (1): 0.9790209790209791

Confusion Matrix:

[[41  2]
 [ 1 70]]
```

The ANN model achieved **97% accuracy** on the test dataset, indicating high predictive performance.

Conclusion

In this project, a breast cancer prediction system was developed using machine learning techniques. The model was implemented using an Artificial Neural Network with the help of the scikit-learn and trained using the Breast Cancer Wisconsin Dataset. This dataset contains important tumor characteristics that help identify whether a tumor is malignant (cancerous) or benign (non-cancerous).

The project followed several important steps such as data collection, data preprocessing, feature scaling, splitting the dataset into training and testing sets, building the Artificial Neural Network model, training the model, and evaluating its performance. The neural network was able to learn complex patterns from the dataset and classify the tumors effectively.

The performance of the model was evaluated using different metrics including accuracy score, F1-score, and confusion matrix. The results showed that the model achieved high accuracy in predicting breast cancer cases, which demonstrates the effectiveness of Artificial Neural Networks for classification problems in the healthcare domain.

Furthermore, the proposed model can be improved in the future by using larger medical datasets, applying advanced deep learning techniques, and integrating real-time medical imaging data. Additional features and more complex neural network architectures could further enhance the prediction accuracy and reliability of the system.

In conclusion, the breast cancer prediction system developed in this project demonstrates the practical application of machine learning in healthcare. Artificial Neural Networks provide an efficient and reliable approach for disease prediction and can play an important role in supporting medical professionals in early diagnosis and improving patient care.

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